

Singelton (View Grades)

```
public class GradeViewerService {

    private static volatile GradeViewerService instance;

    private GradeViewerService() {}

    public static GradeViewerService getInstance() {
        if (instance == null) {
            synchronized (GradeViewerService.class) {
                if (instance == null) {
                    instance = new GradeViewerService();
                }
            }
        }
        return instance;
    }

    public void viewGrades(String studentId) {
        /* Read-only DB query logic for grades */
    }
}
```

Singelton (View Attendance)

```
public class AttendanceViewerService {

    private static volatile AttendanceViewerService instance;

    private AttendanceViewerService() {}

    public static AttendanceViewerService getInstance() {
        if (instance == null) {
            synchronized (AttendanceViewerService.class) {
                if (instance == null) {
                    instance = new AttendanceViewerService();
                }
            }
        }
        return instance;
    }

    public void viewAttendance(String studentId, String date) {
        /* Read-only DB query logic for attendance */
    }
}
```

Factory Method (Announcements)

// Product

```
public interface Announcement {  
    void post(String message);  
}
```

```
public class DashboardAnnouncement implements Announcement {  
    public void post(String message) { /* Dashboard logic */ }  
}
```

```
public class EmailAnnouncement implements Announcement {  
    public void post(String message) { /* Email logic */ }  
}
```

// Creator

```
public abstract class AnnouncementCreator {  
    protected abstract Announcement createAnnouncement();  
  
    public void deliverAnnouncement(String message) {  
        Announcement announcement = createAnnouncement();  
        announcement.post(message);  
    }  
}
```

// Concrete Creators

```
public class DashboardAnnouncementCreator extends AnnouncementCreator {  
    protected Announcement createAnnouncement() {  
        return new DashboardAnnouncement();  
    }  
}
```

```
public class EmailAnnouncementCreator extends AnnouncementCreator {  
    protected Announcement createAnnouncement() {  
        return new EmailAnnouncement();  
    }  
}
```

Abstract Factory (Manage User Accounts)

// Abstract Products

```
public interface UserProfile { void setupProfile(); }
```

```
public interface UserPermissions { void assignPermissions(); }
```

// Concrete Products (Student Family)

```
public class StudentProfile implements UserProfile {
```

```
    public void setupProfile() { /* Student profile logic */ }
```

```
}
```

```
public class StudentPermissions implements UserPermissions {
```

```
    public void assignPermissions() { /* Student permissions logic */ }
```

```
}
```

```
// Concrete Products (Teacher Family)
```

```
public class TeacherProfile implements UserProfile {  
    public void setupProfile() { /* Teacher profile logic */ }  
}
```

```
public class TeacherPermissions implements UserPermissions {  
    public void assignPermissions() { /* Teacher permissions logic */ }  
}
```

```
// Abstract Factory
```

```
public interface UserAccountFactory {  
    UserProfile createProfile();  
    UserPermissions createPermissions();  
}
```

```
// Concrete Factories
```

```
public class StudentAccountFactory implements UserAccountFactory {  
    public UserProfile createProfile() { return new StudentProfile(); }  
    public UserPermissions createPermissions() { return new StudentPermissions(); }  
}
```

```
public class TeacherAccountFactory implements UserAccountFactory {  
    public UserProfile createProfile() { return new TeacherProfile(); }  
    public UserPermissions createPermissions() { return new TeacherPermissions(); }  
}
```


Abstract Factory (Add Student Info)

// Abstract Products

```
public interface StudentForm { void renderForm(); }
```

```
public interface DataValidator { boolean validateInput(String name, int age); }
```

// Concrete Products (Web Family)

```
public class WebStudentForm implements StudentForm {
```

```
    public void renderForm() { /* Web DOM rendering */ }
```

```
}
```

```
public class WebDataValidator implements DataValidator {
```

```
    public boolean validateInput(String name, int age) { /* Server-side validation */ return true; }
```

```
}
```

// Concrete Products (Mobile Family)

```
public class MobileStudentForm implements StudentForm {
```

```
    public void renderForm() { /* Mobile native UI rendering */ }
```

```
}
```

```
public class MobileDataValidator implements DataValidator {
```

```
    public boolean validateInput(String name, int age) { /* On-device validation */ return true; }
```

```
}
```

// Abstract Factory

```
public interface StudentInfoUIFactory {  
    StudentForm createForm();  
    DataValidator createValidator();  
}
```

// Concrete Factories

```
public class WebUIFactory implements StudentInfoUIFactory {  
    public StudentForm createForm() { return new WebStudentForm(); }  
    public DataValidator createValidator() { return new WebDataValidator(); }  
}
```

```
public class MobileUIFactory implements StudentInfoUIFactory {  
    public StudentForm createForm() { return new MobileStudentForm(); }  
    public DataValidator createValidator() { return new MobileDataValidator(); }  
}
```

